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NPS/001/005 Specification for Steelwork, Conductor Fittings, Insulator Fittings and Stay Fittings for Overhead Lines.

1. Purpose

The purpose of this document is to outline the requirements for all steelwork, conductor and insulator fittings and galvanised stay fittings for overhead lines and general construction works for use on the CE Electric UK distribution network.

This document supersedes the following documents, all copies of which should be destroyed.

Ref	Version	Title
DSS/004/001	1.0	Specification for 'D' Irons, Pins and Stay Rods
DSS/004/007	1.0	Specification for Galvanised Insulator Pins
DSS/004/008	1.0	Specification for Overhead Line Steelwork
DSS/004/014	1.0	Specification for Conductors and Stay Fittings
MS-O-35		Specification for Steelwork, Conductor /Insulator Fittings and Stay Fittings for Overhead Line and General Construction Work
NPS/001/005	1.0	Specification for Steelwork, Conductor/Insulator Fittings and Stay Fittings for Overhead Lines and General Construction Work

2. Scope

This specification details the technical requirements for overhead line steelwork and fittings for use on the CE Electric UK distribution network. This includes pole top steelwork, steelwork for ancillary equipment, brackets for terminating low voltage overhead lines and services, stay fittings and conductor/insulator fittings. The majority of items are detailed in national, international and industry standards that are listed within the technical particulars for each type.

3. Technical Specification

3.1 Steelwork

The following general requirements shall apply to the range of steelwork required for the construction of overhead lines on wood poles, buildings and towers. In general items shall comply with ENA Technical Specification 43-95 - Issue 5: 1989.

Unless otherwise specified the steel used in the manufacture of items to this standard shall be to BS EN 10 025 Grade S.275JR.

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Equal and unequal angle sections shall conform to BS EN 10056 Part 1. Channel sections shall conform to BS 4, Part 1. Flat sections shall comply with BS 6722.

Bending of flat straps shall be carried out cold then stress-relieved at 600°C minimum or hot bent before galvanising.

All ferrous materials shall be hot dip galvanised in accordance with BS EN 1461 after fabrication.

Welding shall be carried out in accordance with BS EN 1011 Part 1. After welding and before galvanising, welds shall be thoroughly cleaned to remove slag and splatter, particular attention being paid to the toes of the welds and to the prevention of entrapment of slag and splatter. The preferred method of cleaning welds before galvanising is by sandblasting.

Dimensions are subject to the following tolerances, unless otherwise stated: -

Dimensions up to and including 50 mm - ± 1 mm

Dimensions greater than 50 mm - ± 2 mm

Limiting dimensions of holes shall be within the tolerances or the 'medium fit series' of BS EN 20273 the hole diameters shown on the drawing correspond to the 'minimum diameter' column of the 'medium fit series'.

Fasteners and washers shall be in accordance with ENA TS 43-96.

3.2 Conductor and Insulator Fittings

In general fittings should comply with ENA TS 43-92 and BS 3288 Parts 1 and 2 and shall be: -

- (i) Of adequate strength for the intended application of the fitting and free from defects which would affect the performance of the fitting.
- (ii) Compatible with the conductor material or capable of being used with an intermediate such that there can be no deleterious effect on the conductor or fitting resulting from their association.
- (iii) Not adversely affected in the long term by any coating applied for corrosion or protection.

Ferrous castings shall be manufactured in accordance with BS 3288-1 1997 section 6.1.

Ferrous materials shall be hot dip galvanised in accordance with BS EN 1461 after fabrication.

Fasteners and washers shall be in accordance with ENA TS 43-96

3.3 Stay Fittings

In general fittings should comply with ENA TS 43-91 where applicable. All ferrous materials shall be suitably protected against corrosion such as can occur during service by hot dip galvanisation to the requirements of BS EN 1461. The tube portion of the adjustable stay rod shall be manufactured from heavy gauge steel tube in accordance with BS EN 10255. Threads shall be to BS 3643 course pitch class with a 7H/8g tolerance.

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3.4 Inspection and Testing

3.4.1 General Requirements

All routine and regular sample tests detailed in the Standards referred to in this specification shall be carried out at each stage of manufacture.

Clauses 3.4.1 to 3.4.8 detail the specific testing requirements dependant on the component type. Appendix 1 contains self certification conformance declarations and should be completed for each product group offered.

3.4.2 Steelwork

Steelwork shall be inspected and tested in accordance with BS EN 10 025.

3.4.3 Insulator Set Fittings and Earth Conductor Fittings

Insulator fittings and earth conductor fittings shall be tested and inspected in accordance with BS 3288 – 1 1997 section 6 and BS EN 61284.

3.4.4 Insulator Pins

Insulator pins shall be tested and inspected in accordance with BS 3288 – 1 1997 section 5 and BS EN 61284.

3.4.5 Suspension Clamps

Suspension clamps shall be tested and inspected in accordance with BS 3288 – 1 1997 section 9 and BS EN 61284.

3.4.6 Mechanical Protective Fittings

Mechanical protective fittings shall be tested and inspected in accordance with BS 3288 – 1 1997. section 11 and BS EN 61284.

3.4.7 Electrical Control Fittings (arc protection devices)

Electrical control fittings shall be tested and inspected in accordance with BS 3288 – 1 1997 section 10.

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4. References

4.1 External Documentation

Reference	Title
ENA TS 43-95 Issue 5 1989	Steelwork for Overhead Lines
ENA TS 43-96 Issue 2: 1986	Specification for fasteners and washers for wood poles overhead lines
ENA TS 43-91 Issue 4: 1984	Specification for stay strands and stay fittings for overhead lines
BS EN 10 025	Hot rolled products of structural steel
ENA TS 43-92 Issue 2 1985	Conductor Fittings for Overhead Lines
BS 3288 Part 1: 1997	Insulators and Conductor Fittings for Overhead Power Lines. Performance and general requirements
BS 3288 Part 2: 1990	Insulators and Conductor Fittings for Overhead Power Lines. Specification for a range of fittings
ENA TS 43-91 Issue 5 2004	Stay Strands and Stay Fittings for Overhead Lines
BS EN 61284: 1998	Overhead Lines Requirements and Tests for Fittings
BS EN 10056 – 1: 1999	Specification for structural steel equal and unequal angles. Dimensions
BS 6722: 1986	Recommendations for dimensions of metallic materials
BS EN 1011 Part 1: 1998	Welding. Recommendations for welding of metallic materials. General guidance for arc welding
BS EN 20273: 1992	Fasteners. Clearance holes for bolts and screws
BS EN 10255: 2004	Non-alloy steel tubes suitable for welding and threading. Technical delivery conditions
BS 4 Part 1: 1983	Structural steel sections. Specification for hot-rolled sections
BS 3643: 1981	ISO metric screw threads
BS EN 1461: 1999	Hot Dipped Galvanised Coating on Fabricated Iron or Steel Articles

4.2 Internal documentation

Reference	Title
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5. Definitions

Term	Definition
ENA TS	Energy Networks Association Technical Specification

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6. Authority for issue

6.1 Author

I sign to confirm that I have completed and checked this document and I am satisfied with its content and submit it for approval and authorisation.

		Sign	Date
S Salkeld	Standards Engineer	S Salkeld	25/09/07

6.2 Policy Sponsor

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

		Sign	Date
G Bolden	Purchasing Manager	G Bolden	01/10/07

6.3 Technical Assurance

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

		Sign	Date
G Hammel	Overhead Standards Manager	G Hammel	25/10/07

6.4 DBD Assurance

I sign to confirm that this document has been assured for issue on to the DBD system

		Sign	Date
S Johnson	DBD Administrator	S Johnson	18/10/07

6.5 Approval

Approval is given for the content of this document

		Sign	Date
C Holdsworth	Standards and Assessment Manager	C Holdsworth	01/10/07

6.6 Authorisation

Authorisation is granted for publication of this document

		Sign	Date
C Morgan	Head of Procurement	C Morgan	05/10/07

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Appendix 1 - SELF CERTIFICATION CONFORMANCE DECLARATION

Steelwork, Conductor fittings, insulator fittings and stay fittings shall comply with the latest issues of the relevant international and international standards.

This check sheet identifies the clauses of the standard relevant to steelwork, Conductor fittings, insulator fittings and stay fittings for use on the CE Electric UK distribution network. The manufacturer shall declare conformance or otherwise, clause by clause, using the following levels of conformance declaration codes.

Conformance declaration codes

N/A = Clause is not applicable/ appropriate to the product

Cs1 = The product conforms fully with the requirements of this clause

Cs2 = The product conforms partially with the requirements of this clause

Cs3 = The product does not conform to the requirements of this clause

Cs4 = The product does not currently conform to the requirements of this clause, but the manufacturer proposes to modify and test the product in order to conform.

Instructions for completion

- When Cs1 code is entered no remark is necessary
- When any other code is entered the reason for non-conformance shall be entered
- Prefix each remark with the relevant 'BS EN' 'IEC' or 'ENATS' as appropriate.

Manufacturer:

Product Reference:

Name:

Signature:

Date:

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Appendix 1

Steelwork

- ENATS 43-95 Issue 5: 1989

Clause/Sub-clause	Requirement	Conformance Code	Remarks
3.0	Materials and finish		
4.0	Welding		
5.0	Tolerances		

- BS EN 1461: 1999

Clause/Sub-clause	Requirement	Conformance Code	Remarks
5.0	Sampling		
6.1	Appearance		
6.2	Thickness		
6.3	Renovation		
6.4	Adhesion		
6.5	Acceptance criteria		

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Steelwork

- BS EN 10025 – 3: 2004 Type S.275JR

Clause/Sub-clause	Requirement	Conformance Code	Remarks
4.0	Classification and designation		
6.0	Manufacturing process		
7.1	General requirements		
7.2	Chemical composition		
7.3	Mechanical properties		
7.4	Technological properties		
7.5	Surface properties		
7.6	Internal soundness		
8.0	Inspection		
10.0	Test methods		

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Conductor and Insulator Fittings

- BS 3288 Part 1: 1997

Clause/Sub-clause	Requirement	Conformance Code	Remarks
4.0	General Requirements		
5.0	Requirements for insulator pins		
6.0	Insulator set and earth conductor fittings		
9.0	Suspension clamps		
10.0	Electrical control fittings (arc control devices)		

- BS 3288 Part 2: 1990

Clause/Sub-clause	Requirement	Conformance Code	Remarks
3.0	Dimensions and tolerances		
4.0	Insulator pins		
7.0	Insulator set fittings		

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Stay Fittings

- ENA TS 43-91 Issue 4: 1984

Clause/Sub-clause	Requirement	Conformance Code	Remarks
4.1.2	Ferrous components		
4.2.1	General requirements for Stay Assemblies		
4.2.13	Stay thimbles		
5.1	Stay rod material requirements		
5.2	Test requirements		
8.0	Stay attachments and thimbles		